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JANUARY 23, 1957
SUPERSEDING
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December 20, 1954

FEDERAL SPECIFICATION

RODENTICIDE, BAIT, ANTICOAGULANT

This specification was approved by the Commissioner, Federal Supply Service, General Services Administration, for the use of all Federal agencies.

1. SCOPE AND CLASSIFICATION

1.1 Scope.—Rodenticide, Bait, Anticoagulant, covered by this specification, is suitable for the control of commensal rats and mice.

1.2 Classification.—Rodenticide, Bait, Anticoagulant, covered by this specification shall be of but one type and grade, as hereinafter specified.

2. APPLICABLE SPECIFICATIONS, STANDARDS, AND OTHER PUBLICATIONS

2.1 Specifications. — The following specifications and standards, of the issues in effect on date of invitation for bids, form a part of this specification:

Federal Specifications:

N–O–41—Oats; Rolled.

RR–S–366—Sieves, Standard for Testing Purposes.

PPP–C–96—Cans, Metal, 28 Gage and Lighter.

PPP–P–31 — Preservation, Packaging, Packing, and Marking of Material; and Shipping Instructions (Domestic Shipment and Storage).

(Activities outside the Federal Government may obtain copies of Federal Specification and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U. S. Government Printing Office, Washington 25, D. C.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and the Standards from established distribution points in their agencies.)

Military Standards:

MIL–STD–105 — Sampling Procedures and Tables for Inspection by Attributes.

MIL–STD–129 — Marking for Shipment and Storage.

(Copies of Military specifications required by contractors in connection with specific procurement function should be obtained from the procuring agency or as directed by the contracting officer.)

2.2 Other publications. — The following publications, of the issues in effect on date of invitations for bids, unless otherwise stated, form a part of this specification:

Governmental:

Federal Insecticide, Fungicide and Rodenticide Act of 1947, as amended.

(Copies of the Federal Insecticide, Fungicide, and Rodenticide Act may be obtained upon application accompanied by money order, coupon, or cash, to the Superintendent of Documents, Government Printing Office, Washington 25, D. C. Prices may be obtained from the Superintendent of Documents.)

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FSC Class 6840

Nongovernmental:

Pharmacopoeial Conventions Publication:

Pharmacopoeia of the United States.

(The Pharmacopoeia of the United States is published by the Mack Printing Company, Easton, Pennsylvania.)

Association of Official Agricultural Chemists Publication:

Official Methods of Analysis of the Association of Official Agricultural Chemists.

(Official Methods of Analysis of the Association of Official Agricultural Chemists is published by the Association of Official Agricultural Chemists, P. O. Box 540, Benjamin Franklin Station, Washington 4, D. C.)

Consolidated Classification Committee Publication:

Consolidated Freight Classification Rules.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Chicago Union Station, Chicago 6, Ill.)

3. REQUIREMENTS

3.1 Material.—The rodenticide shall consist of a chemical anticoagulant which may be warfarin, or pival, or fumarin suitably compounded with petrolatum, sucrose, rolled oats and a dye, the admixture conforming to the compositional requirements given in table I.

TABLE I.—*Composition*

Ingredient	Percent	
	Minimum	Maximum
Anticoagulant	0.025	0.028
Liquid petrolatum	4.8	5.2
Sucrose	4.8	5.2
Dye	0.05	
Breakfast rolled oats	Remainder	

3.1.1 *Warfarin.* — Warfarin shall be the pure chemical 3– (alpha – acetonylbenzyl) – 4 – hydroxycoumarin, and the percentage shown in the ingredient statement shall represent pure warfarin. This warfarin shall be derived from a technical product which shall be a fine white or creamy white powder, melting at 159°–163°C., and containing not less than 98 percent warfarin. It shall be free of objectional odor or taste which might limit its suitability for the proposed end use (see 3.2 and **4.4.2**).

3.1.2 *Pival.*—Pival shall be the pure chemical 2 – pivalyl 1, 3 – indandione, and the percentage shown in the ingredient statement shall represent pure pival. This pival shall be derived from a technical product which shall be a fine yellow or brownish-yellow powder, melting at 107°–110°C., and containing not less than 98 percent pival. It shall be free of objectionable odor or taste which might limit its suitability for the proposed end use (see 3.2 and 4.4.1).

3.1.3 *Fumarin.* — Fumarin shall be the pure chemical 3– (alpha – acetonylfurfuryl) – 4 – hydroxycoumarin, and the percentage shown in the ingredient statement shall represent pure fumarin. This fumarin shall be derived from a technical product which shall be a fine white or creamy white powder, melting at 120°–123°C., and containing not less than 98 percent fumarin. It shall be free of objectionable odor or taste which might limit its suitability for the proposed end use (see 3.2).

3.1.4 *Liquid petrolatum.*—The liquid petrolatum shall be medicinal grade mineral oil, with a specific gravity of not less than 0.83 nor more than 0.86.

3.1.5 *Sucrose.*—The sucrose shall conform to the requirements of the Pharmacopoeia of the United States.

3.1.6 *Dye.*—The dye shall be an oil-soluble, brilliant red dye, approved by the Fish and Wildlife Service. It shall be equivalent to

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Flaming Red Dye No. G305, manufactured by the Ox Color Works, New York, N. Y.

3.1.7 *Rolled oats.*—The rolled oats shall conform to the requirements for type I, Regular, Rolled Oats, as specified in Federal Specification N–O–41.

3.2 Certificate of compliance. — For each delivery of finished rodenticide, the manufacturer shall submit a certified copy of the production record, showing:

(a) Certified copies of analyses of chemical anticoagulant, liquid petrolatum, sucrose, and rolled oats used in formulation.

(b) Exact weights of all ingredients used.

(c) Time spent in mixing or blending.

(d) Weight of finished product.

3.3 Particle size.—Not more than 30 percent of the finished rodenticide shall pass through a No. 20 U. S. Standard woven-wire cloth sieve, when tested as described in 4.4.3.

3.4 Moisture. — The finished rodenticide shall not lose more than 12 percent of its weight when tested as described in 4.4.4.

3.5 Performance.—The toxicity and performance of the finished rodenticide shall be such that not more than 20 percent of the test animals shall survive when the product is tested as described in 4.4.7.

3.6 Federal Insecticide, Fungicide and Rodenticide Act.—The finished rodenticide shall comply with the applicable provisions of the Federal Insecticide, Fungicide and Rodenticide Act (see 6.5).

3.7 Instruction sheet. — Twenty-four printed or mimeographed instruction sheets containing the following information, each placed in a separate envelope marked "Important Instructions", shall be enclosed in each intermediate container:

INSTRUCTIONS FOR USING RODENTICIDE, BAIT, ANTICOAGULANT

1. Do not breathe the dust or swallow any of the poison. Do not smoke or eat while handling the poison. Keep personnel contact with the rodenticide at a minimum.

2. Keep all equipment labeled plainly, wash it thoroughly after use, and do not employ it for other purposes.

3. Wash hands thoroughly after handling the poisoned bait.

METHOD FOR USE IN CONTROL OF RATS AND MICE

Rodenticides, containing warfarin, pival, and fumarin reduce the blood coagulation rate and death ensues in most cases from internal hemorrhages. At concentrations recommended for rat and mouse control, warfarin, pival, and fumarin apparently are not detected, or, if so, their presence in food baits are not objectionable to rats and mice. It has been a frequent observation, both in the laboratory and in the field, that rats continue to attempt to feed on the baits when they are under the influence of the drug to such an extent that they lack the strength to stand.

This rodenticide will not give effective control when applied in a single dose, but must be ingested several times (usually 5 or more) on successive days. These feedings need not be on consecutive days but should occur within a 10-14 day interval.

The control of rats, including roof rats and house mice, when the concentration of warfarin, pival, or fumarin in the bait is 0.025 percent (0.25 mg. of warfarin, pival, and fumarin per gram of bait), has been obtained.

BAIT EXPOSURE

A. *Rats*

Exposure of baits should be made by establishing an adequate number of protected bait stations. This can be accomplished by placing open trays of bait in rooms or areas

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about premises where domestic animals and small children can be excluded or through the use of properly constructed bait boxes. In many instances baits can be protected by improvised means with materials available on the premises. This is especially true where it is possible to employ scrap lumber for this purpose.

Feeding stations should be inspected at frequent intervals for the purpose of replenishing the bait supply where necessary. The frequency of inspections and size of bait placements will, of course, depend upon the degree of infestation encountered and in some instances daily servicing of the bait stations may be required during the first 4-5 days of a treatment. Bait consumption and rat activity usually drop markedly about 5 days after a rat population begins to feed on baits. Control is usually achieved in 10-14 days after feeding begins. Baits should be kept dry during use to maintain maximum acceptability.

It is often desirable to maintain properly protected bait stations with semi-permanent baits after control has been attained on premises which are particularly vulnerable to re-invasion.

Four to eight ounces of bait are suggested as the amount to be placed in each feeding station. Subsequent inspection will determine the correct amount to be added to each station.

B. *Mice*

In general, the same exposure technique should be used in employing baits for the control of mice as outlined for rat control; the principal difference, however, being that a larger number of placements should be made in areas in which the mice are known to feed. Those feeding stations may in some cases consist of only tablespoonful quantities.

Inspections should be made frequently enough throughout a treatment to maintain a continuous supply of bait available at each placement.

PRECAUTIONS

Warfarin, pival, and fumarin probably will kill any warm-blooded animal if consumed in the proper dosage over an adequate period of time. However, the concentration used in rodent control requires that on the basis of a single dose ingestion, relatively large quantities of the bait be consumed before domestic animals are seriously endangered. On the other hand, if daily feeding on warfarin, pival, and fumarin by domestic animals is allowed, much smaller amounts may be fatal. In one of the 521 exposures analyzed, a cat was said to have died from hemorrhage as a result of preying on rats and mice killed from feeding on warfarin, pival, and fumarin baits. In view of this possibility, it is recommended that, where possible, dead rats and mice killed through the ingestion of these rodenticides be disposed of in a manner such that cats and dogs will not have access to them.

ANTIDOTE

Warfarin, pival, and fumarin are anticoagulant chemical, which, if taken accidentally by humans or domestic animals, may increase the clotting time of the blood and cause hemorrhage. Such hemorrhages may be either internal or external and often hematoma are visible under the skin of animals affected with these substances.

In the case of accidental ingestion by humans, vomiting should be induced at once and a physician called. Treatment by the physician should include transfusion with whole blood and intravenous and oral administration of Vitamin K preparations as in the case of hemorrhage caused by Dicumarol.

3.8 Workmanship.—The method employed in making the rodenticide shall be in accordance with best commercial practice. The product shall be clean, uniformly blended, and free of defects which might impair its utility.

4. SAMPLING, INSPECTION, AND TEST PROCEDURES

4.1 Sampling for lot acceptance.

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4.1.1 *Inspection lot.*—For purposes of sampling, a lot shall consist of all rodenticide manufactured as one batch, and offered for delivery at one time.

4.1.2 *Sampling for inspection of filled containers.*—A random sample of filled containers shall be taken from each lot offered for inspection in accordance with Military Standard MIL–STD–105 at inspection level I, and acceptable quality level (A.Q.L.) = 2.5 percent defective to verify compliance with all stipulation of this specification regarding fill, closure, marking and other requirements not involving tests.

4.1.3 *Sampling for tests.*—From each inspection lot offered for testing, enough containers shall be taken at random to form two separate five-pound specimens. The containers shall be sealed, marked, and forwarded to the testing laboratory designated by the bureau or agency concerned.

4.1.3.1 *Components.*—The following samples of components shall be taken and forwarded to the Government testing laboratory:

Warfarin, Pival or Fumarin (as applicable)	4 ounces
Dye	1 ounce
Liquid petrolatum	4 ounces
Sucrose	1 pound
Rolled oats	2 pounds

4.2 Inspection.

4.2.1 *Inspection of filled containers.*—Each sample filled container selected in accordance with 4.1.2 shall be examined for defects of the container and the closure, for evidence of leakage, and for unsatisfactory markings; each sample filled container shall also be weighed to determine the amount of the contents. Any container in the sample, having one or more defects, or under required fill, shall be rejected and, if the number of defective containers in any sample exceeds the acceptance number for the appropriate sampling plan of Military Standard MIL–STD–105, the lot represented by the sample shall be rejected. Rejected lots may be resubmitted for acceptance tests provided that the contractor has removed or repaired all nonconforming containers.

4.3 Lot acceptance tests. — The sample specimens selected in accordance with 4.1.3 shall be subjected separately to the tests specified in 4.4. If either specimen fails one or more of these tests, the lot shall be rejected. Rejected lots may be resubmitted for acceptance tests provided the contractor has removed or reworked all nonconforming products.

4.4 Test procedures.

4.4.1 *Pival determination.* — Accurately weigh 0.1 gm. pival into a 250 ml. Erlenmeyer flask and add 40 ml. of ethanol (95 percent). Heat on steam bath for 10 minutes with constant stirring or swirling. Pour into a 100 ml. volumetric flask and make to volume at 25°C. Transfer a 10 ml. aliquot into a 1000 ml. volumetric flask and adjust the volume to 1000 ml. at 25°C. Pipette a sufficient amount (approximately 3 ml.) of the above solution into a 1 cm. quartz cuvette, and determine absorbency (As) at 283 Mu in a Beckman spectrophotometer, model DU or equivalent, using a blank made up of 95 percent ethanol. The percentage of pival is calculated by the following formula:

$$\text{Pival, percent} = \frac{\text{As} \times 0.2340}{\text{Weight of sample (gm.)}}$$

4.4.2 *Warfarin.*

4.4.2.1 *Determination.*—Weigh 0.6 gm. of warfarin into a 100-ml. volumetric flask and make up to volume at 25°C. Place a 1-ml. aliquot in a 250-ml. glass-stoppered flask, and add 100 ml. of ethylene dichloride with a pipette. Shake on a shaking machine for at least 10 minutes. Pipette 2 ml. of the ethylene dichloride solution into a glass stoppered cylinder, add 10 ml. of 1 percent NaOH solution with a pipette and shake for 1 minute by hand. Prepare a blank solution

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in the same manner, using 2 ml. ethylene dichloride. Decant the sodium hydroxide layer into a centrifuge tube, and centrifuge until clear. Pipette sufficient amount of the alkali layer (approx. 3 ml. into a quartz cuvette (1 cm. cell) and determine the optical density at 308 mu and slit opening of about 0.86 mm, using a Beckman spectrophotometer, model DU or equivalent.

4.4.2.2 *Sample calculation.* — The optical density of a solution of 1 mg. warfarin in 100 ml. of 1 percent sodium hydroxide is found to be 0.463, using the above procedure. If D is the optical density:

$$\text{Warfarin, percent} = \frac{D(10/100)\ (100)}{0.463\ (1000)\ (0.6)\ (2/50)} \text{ or } = D(0.9)$$

4.4.3 *Sieve test.*—Transfer 100 grams of the finished rodenticide to a No. 20 U. S. Standard sieve (conforming to Federal Specification RR–S–366) and screen in a Ro-tap or equivalent machine for 2 minutes. Weigh the material which passes through the sieve and calculate the percentage.

4.4.4 *Moisture.*—Accurately weigh approximately 2 grams of the finished rodenticide into a tared moisture dish (Ni, Pt, or Al) and dry for 10 hours under atmospheric pressure at a temperature of 100°C. (± 2°C); cool in desiccator and weigh. Dry again for one hour, or until change in weight is not more than 2 mg. Report loss in weight as moisture.

4.4.5 *Liquid petrolatum.*—Accurately weigh approximately 100 grams of the finished rodenticide, ground to pass a 20 mesh screen, prepare ether extract and determine percentage of mineral oil according to procedures prescribed in Methods of Analysis of the Association of Official Agricultural Chemists.

4.4.6 *Sucrose.*—Accurately weigh sample of finished rodenticide powder and determine sucrose content according to procedures described in Methods of Analysis of the Association of Official Agricultural Chemists.

4.4.7 *Performance.*—The suitability of the finished rodenticide for the proposed end use shall be determined according to the following procedure:

4.4.7.1 *Materials.*

4.4.7.1.1 *Bio-assay animals.*—The bio-assay animals shall consist of mature, healthy, male albino rats of a Wistar strain weighing between 150 and 250 grams.

4.4.7.1.2 *Control diet.*—The control diet shall consist of pelleted laboratory chow, or dog biscuits, equivalent to that commonly employed for feeding of laboratory rats.

4.4.7.2 *Procedure.*

4.4.7.2.1 *Preliminary treatment of bio-assay animals.*—All animals used in this test shall have been maintained on the control diet for not less than two weeks prior to the beginning of the assay. Twenty-four hours prior to the start of the test period, thirty mature, healthy, male albino rats are to be deprived of all food for a period of 24 hours, then weighed and transferred to individual feeding cages

4.4.7.2.2 *Bio-assay procedure.*—Doses are computed at the level of 5 mg. warfarin, pival or fumarin per kilogram rat body weight, and the necessary quantity of food calculated for each animal each day for 8 consecutive days. Test doses are to be presented in the afternoon and the rats are to be given access to the control diet during a 20-minute period the following morning. This is to prevent a condition of starvation which would occur if nothing but the test material were administered during the 8-day interval. At the end of the 8-day test period, the rats are to be given ad libitum access to the control diet and observed for a period of 10 days following the ingestion of the final dose of poison. The temperature of the room should be kept between 65° and 80°F., and the animals are to be given ad libitum access to water at all times. If 80 percent or more die, the rodenticide shall be deemed acceptable.

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5. PREPARATION FOR DELIVERY

(For definitions and levels see 6.1).

5.1 Packaging

5.1.1 *Levels A and B.* — The rodenticide shall be packaged in quantities of 2 ounces, 1 pound, or 3 pounds as specified in the contract or order, in round hermetically sealed open-top cans conforming to type I of Federal Specification PPP–C–96, Plan B coating. The product shall be compressed to a density of at least 1 ounce for each 2.6 cubic inches based on the net inside packing space of the can.

5.1.2 *Level C.*—Rodenticide shall be packaged in accordance with the manufacturer's commercial practice.

5.2 Packing.

5.2.1 *Level A.*—Rodenticide shall be packaged in accordance with the overseas requirements of Federal Specification PPP–C–96 and appendix thereto.

5.2.2 *Level B.*—Rodenticide shall be packed in accordance with the domestic requirements of Federal Specification PPP–C–96 and appendix thereto.

5.2.3 *Level C.*—Rodenticide shall be packed in a manner to insure safe delivery and acceptance at destination. Containers shall comply with the Consolidated Freight Classification Rules or other carrier regulations as applicable to the mode of transportation.

5.3 Marking.

5.3.1 *Civil agencies.*—In addition to the marking specified in 5.3.3 and any special marking required by the contract or order, marking for shipment shall be in accordance with Federal Specification PPP–P–31.

5.3.2 *Military.*—In addition to the marking specified in 5.3.3 and any special marking required by the contract or order, marking for shipment shall be in accordance with Military Standard MIL–STD–129.

5.3.3 *Labeling.* — The shipping container and each individual package shall be labeled to comply with the provisions of the Federal Insecticide, Fungicide, and Rodenticide Act of 1947, and shall be clearly and legibly marked with the following applicable label data:

RODENTICIDE, BAIT
ANTICOAGULANT (1)

Stock No. ------

	Percent
Active Ingredients (2)	0.025
Inert Ingredients	99.975

(1) Insert either WARFARIN, PIVAL, or FUMARIN

(2) Insert either Warfarin, 3- (alpha-acetonylbenzyl) - 4 - hydroxycoumarin; 2 - pivalyl - 1, 3 - indandione; or 3 - (alpha - acetonyl - furfuryl) - 4 - hydroxycoumarin.

Directions.—Place 4 to 8 ounces of Rodenticide, Bait, Anticoagulant in bait stations. Place so as to protect children, pets, and livestock in locations where rats or mice feed. Replenish baits as necessary, and maintain stations for not less than 12 days. (Baiting may be continued indefinitely, and increased time of exposure of baits may result in better control of rodents.) If bait molds, decomposes, or becomes sour, replace with fresh material.

CAUTION.[1]—Keep away from humans, domestic animals, and pets. If swallowed by humans, domestic animals, or pets, this material may reduce the clotting ability of the blood and cause bleeding. In such case, intravenous and oral administration of Vitamin K combined with blood transfusions are indicated as in the case of hemorrhage caused by overdoses by bis-hydroxycoumarin.

[1] This caution paragraph is optional.

6. NOTES

6.1 Federal Standard 102 should be referred to for definitions and applications of

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the various levels of packaging protection for supplies and equipment.

6.2 Procurement document should specify the following:

Whether packaging and packing shall be by level A, B, or C (see section 5).

6.3 **Intended use.**—Rodenticide is intended for use in the control of commensal rats and mice.

6.4 **Approved laboratory.** — The Wildlife Research Laboratory, U. S. Fish and Wildlife Service, Denver, Colorado, is an approved laboratory for testing the rodenticide to determine compliance with the requirements of this specification.

6.5 The Federal Insecticide, Fungicide, and Rodenticide Act of 1947 is under the jurisdiction of the Department of Agriculture, Pesticide Regulation Section, Plant Pest Control Branch, of the Agricultural Research Service, Washington 25, D. C., to which Agency all matters pertaining to registration should be addressed.

6.6 **Transportation description.** — Transportation description applicable to this item is:

Insecticides, insect repellents or vermin exterminators not otherwise indexed by name, other than agricultural insecticides.

Carload minimum weight 36,000 pounds.

Truckload minimum weight 30,000 pounds.

Patent notice.—When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data, is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use or sell any patented invention that may in any way be related thereto.

MILITARY INTERESTS:
Army—Q E M
Navy—Sh Y MC
Air Force.

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